

Day 20 - Lecture 20.4: Solving Foundation Probability Practice Problems

 Python 3.10+

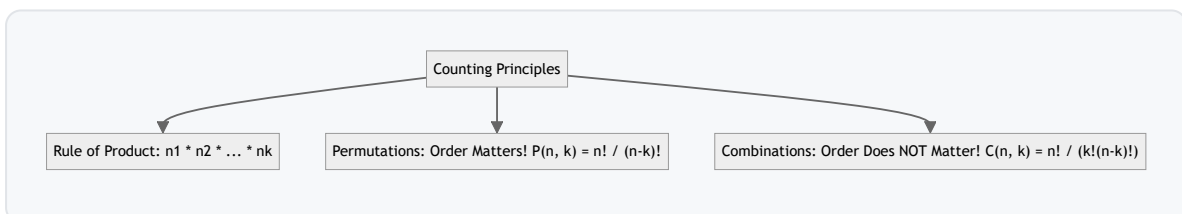
 Jupyter Notebook

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1. Combinatorics & Counting Principles in Probability

Calculating probabilities in discrete sample spaces requires the fundamental principles of combinatorics: the **Rule of Product**, **Permutations**, and **Combinations**.



Mathematical Definitions:

Permutations (Order is Distinct):

The number of ordered sequences of k distinct elements chosen from an n -element set:

$$P(n, k) = \frac{n!}{(n-k)!}$$

Combinations (Order is Irrelevant):

The number of subsets of size k chosen from an n -element set:

$$C(n, k) = \binom{n}{k} = \frac{n!}{k!(n-k)!}$$

2. Classic Problem Formulations & Analytical Solutions

Problem 1: Rolling Two Dice (Sum = 7 or 8)

- Sample space size: $|\mathcal{S}| = 6 \times 6 = 36$.
- Event A (Sum = 7): $(1, 6), (2, 5), (3, 4), (4, 3), (5, 2), (6, 1) \implies |A| = 6$.
- Event B (Sum = 8): $(2, 6), (3, 5), (4, 4), (5, 3), (6, 2) \implies |B| = 5$.
- Because $A \cap B = \emptyset$ (a sum cannot be both 7 and 8 simultaneously):

$$P(\text{Sum} \in \{7, 8\}) = \frac{6 + 5}{36} = \frac{11}{36} \approx 0.3056$$

Problem 2: Card Drawing Without Replacement

From a standard 52-card deck, draw 2 cards sequentially without replacement. What is the probability that both are Kings?

- Total 2-card hands: $\binom{52}{2} = \frac{52 \times 51}{2} = 1326$.
- Favorable 2-King hands: $\binom{4}{2} = \frac{4 \times 3}{2} = 6$.

$$P(\text{Both Kings}) = \frac{6}{1326} = \frac{1}{221} \approx 0.004525$$

3. Python Implementation & Verification

```
import math
from itertools import product, combinations

# Problem 1: Two Dice Sum Simulation
die = list(range(1, 7))
sample_space_dice = list(product(die, die))
favorable_dice = [pair for pair in sample_space_dice if sum(pair) in (7, 8)]
p_dice_analytical = len(favorable_dice) / len(sample_space_dice)

print(f"Problem 1: |S| = {len(sample_space_dice)}, |Favorable| = {len(favorable_dice)}")
print(f"P(Sum 7 or 8) = {p_dice_analytical:.4f} (Exact: 11/36)")

# Problem 2: Combinatorial Poker Hand
deck_size = 52
kings_in_deck = 4
total_hands = math.comb(deck_size, 2)
king_hands = math.comb(kings_in_deck, 2)
p_two_kings = king_hands / total_hands

print(f"Problem 2: Total 2-Card Hands = {total_hands}, 2-King Hands = {king_hands}")
print(f"P(Both Kings) = {p_two_kings:.6f} (Exact: 1/221 = {1/221:.6f})")
```

4. Key Takeaways & Interview Points

- **Permutations vs. Combinations:** If the order of assignment affects the outcome (e.g., passwords, rankings), use permutations; if only group membership matters (e.g., committee, feature subsets), use combinations.
- **Denominator Discipline:** In conditional or sequential draws, always check whether replacement alters the sample space cardinality $|\mathcal{S}|$.